

Q1. What is the main difference between embedded system and VLSI, and their applications?

Sankar
Tuticorin

A1. An embedded system is a system dedicated to a particular real-time application, with a pre-programmed microcontroller or microprocessor in it. It has a computer as one of its components but it itself is not a general-purpose computer. Embedded system is task-specific. Hardware and software are optimised and are part of a larger system.

Embedded system responds to environment by using sensors. It monitors and controls external environment without human intervention.

Common characteristics of embedded systems are that these are single-functioned, i.e., execute a single program repeatedly. They continuously interact with environment, react to changes and compute results in real-time. Their range varies from handheld portable devices, such as digital watches and MP3 players, to large stationary installations like traffic-lights or the systems controlling nuclear power plants. Other examples are personal digital assistants, mobile phones, video game consoles, digital cameras, DVD players, GPS receivers, microwave ovens and washing machines. However, your desktop computer is not an embedded system.

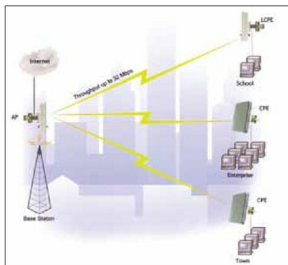
Very large scale integration (VLSI) is a field that involves packing more logic devices into smaller areas. It is a technology that enables us to pack a large number of circuits in a few millimetres space. This has opened up a big opportunity to do things that were not possible before. VLSI circuits are everywhere—in computers, cars, digital cameras, cellphones, and the list goes on.

Relation between embedded systems and VLSI is that the former has devices, may be number of them, made through VLSI.

Q2. I want to know about service range and coverage area of WiMAX system.

Venkatatrishma Reddy
Through e-mail

A2. WiMAX is a wireless metropolitan area network (MAN) technology that can connect IEEE 802.11 (WiFi) hotspots with each other and to



WiMAX system

other parts of the Internet. It provides a wireless alternative to cable and DSL for broadband access. IEEE 802.16 provides up to 50 km of linear service area range and connectivity between users without a direct line of sight.

A WiMAX system has a WiMAX tower—similar in concept to a cell-phone tower—that can provide coverage to a very large area (as big as 8000 square kilometres). Transmitted signals are received by a WiMAX receiver through a very small antenna. A WiMAX tower station can connect directly to the Internet using a high-bandwidth wired connection. It can also be connected to another WiMAX tower using a line-of-sight microwave link.

Q3. How can I use a positive voltage regulator for a battery charging circuit?

U. Sreehari
Hyderabad

A3. You can design a power

supply circuit capable of providing about 15V regulated output, which can provide current up to 2A. Voltage regulators of L78500 series will meet the requirement. The L78500 series of 3-terminal positive regulators is available in TO-220 and TO-3 packages with several fixed output voltages, making it useful in a wide range of applications.

These regulators provide local on-card regulation, eliminating the distribution problems associated with single-point regulation. Each type employs internal current-limiting, thermal shut-down and safe-area protection, making it essentially indestructible. If adequate heat sinking is provided, these can deliver over 2A output current. Although designed primarily as fixed voltage regulators, these devices can be used with external

components to obtain adjustable voltages and currents.

For your application, the power supply output should be more than 12V to charge a 12V battery. So voltage regulator L78515 would be the ideal choice, as it is capable of providing a 2A current. The regulator requires 35V at its input terminal. Calculating backwards, you will have to select the step-down transformer with secondary voltage of about 35V, capable of supplying 2A current. Diodes, which are to be used as rectifier diodes in full-wave mode, should also be capable of handling 2A current. You may also like to add circuits for protection against short circuit and over-charging.

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